$$x(h) = x(0) + h\,v\left(\frac{h}{2}\right) \qquad (2.9)$$

$$x(h) = x(0) + h\left(v\left(-\frac{h}{2}\right) + h\,F(x(0),0)\right) \qquad (2.8)$$

$$x(h) = x(0) + h\,v\left(-\frac{h}{2}\right) + h^2 F(x(0),0)$$

$$\cancel{x(h) = x(0) +}\ v\left(-\frac{h}{2}\right) = \frac{x(0) - x(-h)}{h} + O(h^2) \qquad (2.7)$$

$$x(h) = x(0) + \cancel{x(0) - x(-h)} + h\,O(h^2) + h^2 F(x(0),0)$$

$$x(h) = 2x(0) - x(-h) + h^2 F(x(0),0) + O(h^3)$$